Capacitors and Inductors

Video 10

Electrical Science

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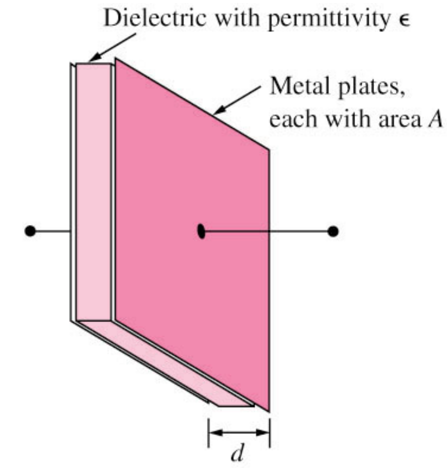
Capacitors and Inductors

- Capacitors
- Series and Parallel Capacitors
- Inductors
- Series and Parallel Inductors

Capacitors



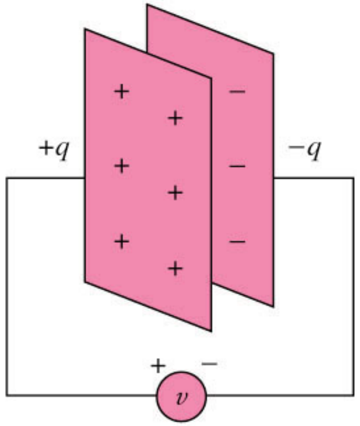
- A **capacitor** consists of two conducting plates separated by an insulator (or dielectric).

- Where ε is the permittivity of the dielectric material between the plates, A is the surface area of each plate, d is the distance between the plates.

$$C = \frac{\varepsilon A}{d}$$

- A capacitor is a passive element designed to **store energy** in its **electric field.**

$$q = C\,v$$

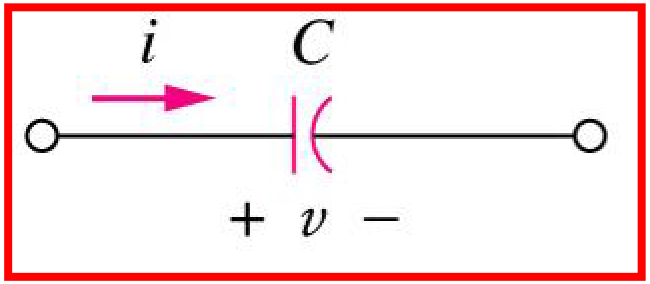


- **Capacitance** *C* is the ratio of the charge *q* on one plate of a capacitor to the voltage difference *v* between the two plates, measured in farads (F).

$$C = q/v$$

Capacitor Current and Voltage

- If *i* is flowing into the +ve terminal of C
  - Charging => *i* is +ve
  - Discharging => *i* is −ve



- The current-voltage relationship of capacitor

$$i = C\frac{dv}{dt}$$

$$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0)$$

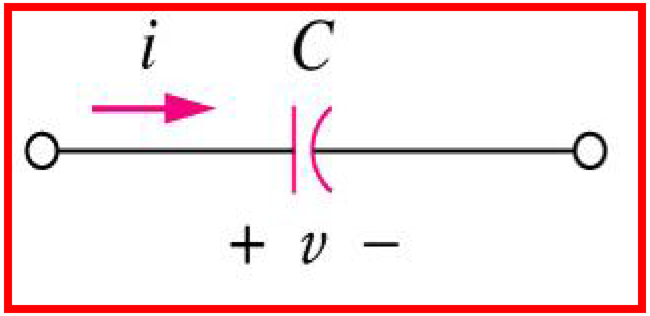
- The energy stored in the capacitor is $$w = \frac{1}{2}Cv^2$$

- A capacitor is an **open circuit** to dc (dv/dt = 0).

Capacitors

Example 1

The current through a 100-μF capacitor is $i(t) = 50 \sin(120\pi t)$ mA. Calculate the voltage across it at t =1 ms and t = 5 ms. Take $v(0)$ =0.

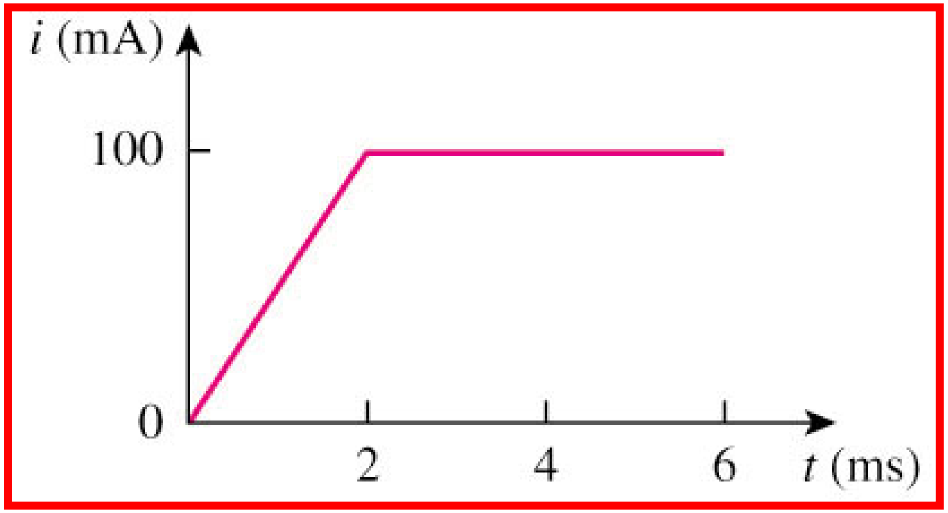


Answer: v(1ms) = 93.14mV   v(5ms) = 1.7361V

Capacitors

Example 2

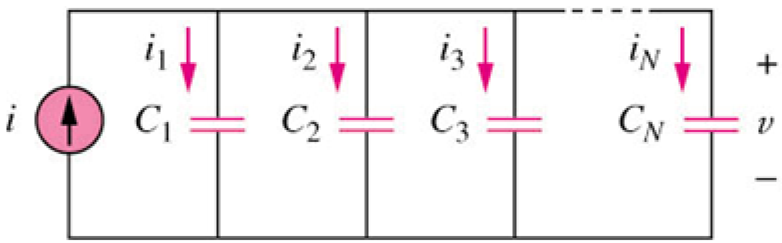
An initially uncharged 1-mF capacitor has the current shown below across it. Calculate the voltage across it at t = 2 ms and t = 5 ms.

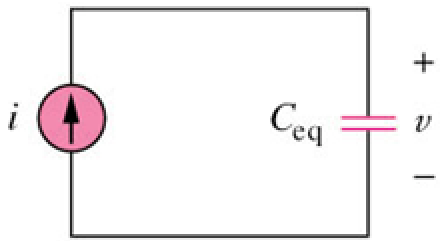


Answer: *v(2ms)* = 100 mV *v(5ms)* = 400 mV

Parallel Capacitors

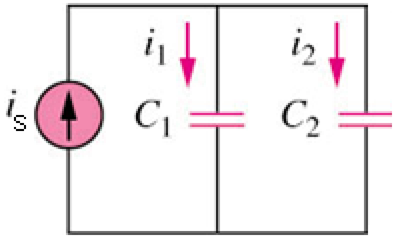
- The equivalent capacitance of *N* **parallel-connected** capacitors is the sum of the individual capacitances.





$$C_{eq} = C_1 + C_2 + \ldots + C_N$$

- Current Divider Rule

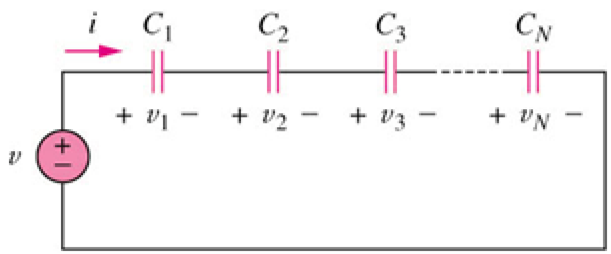


$$i_1 = \frac{C_1}{C_1 + C_2} i_s$$

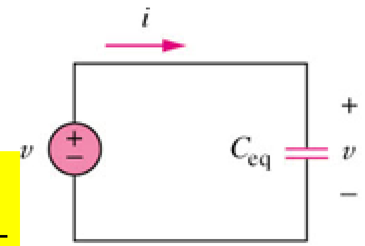
$$i_2 = \frac{C_2}{C_1 + C_2} i_s$$

Series Capacitors

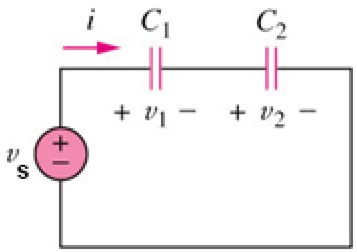
- The equivalent capacitance of *N* **series-connected** capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_N}$$



- Voltage Divider Rule



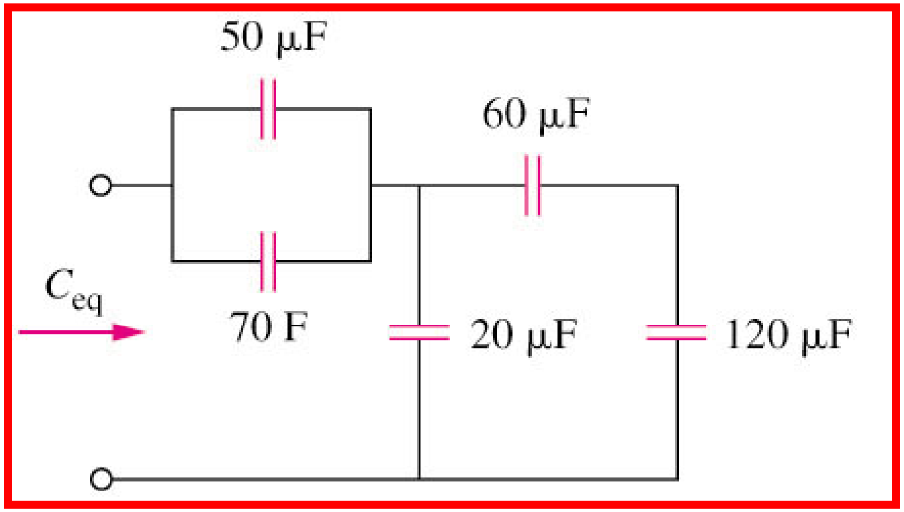
$$v_1 = \frac{C_2}{C_1 + C_2} v_s$$

$$v_2 = \frac{C_1}{C_1 + C_2} v_s$$

Series and Parallel Capacitors

Example 3

Find the equivalent capacitance seen at the terminals of the circuit in the circuit shown below:

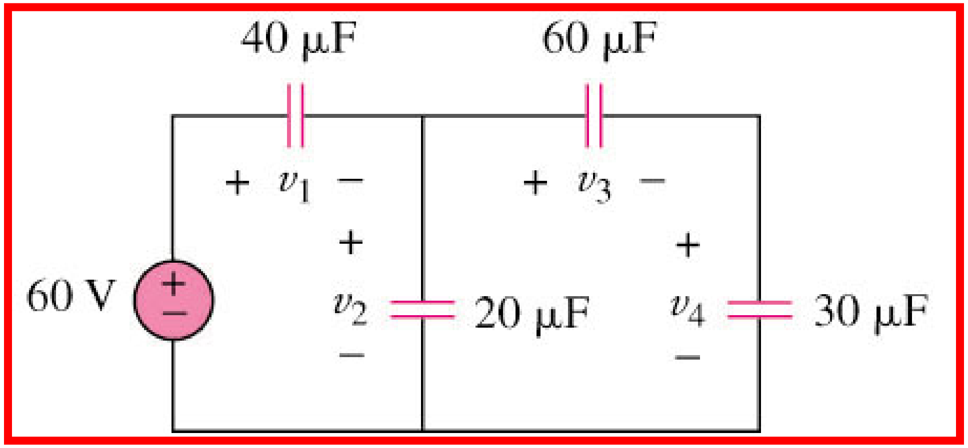


***Answer:* C_{eq} = 40μF**

Series and Parallel Capacitors

Example 4

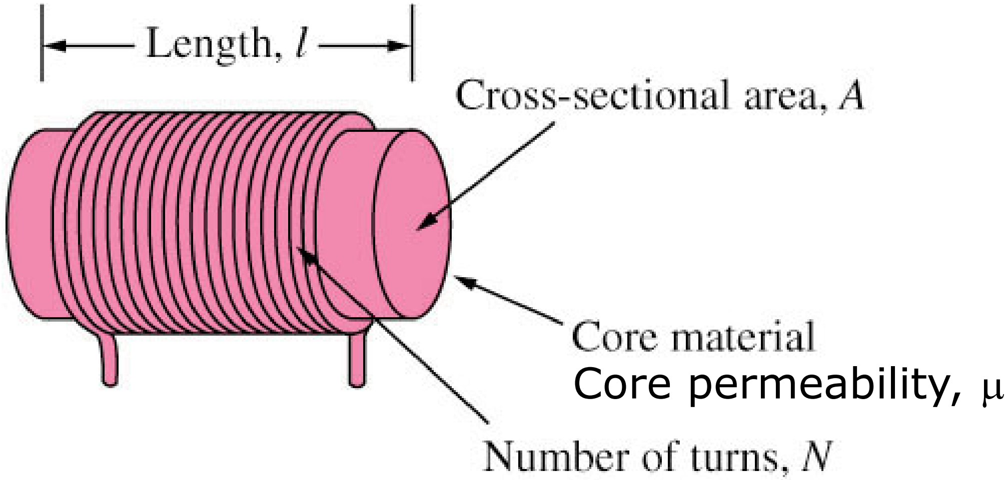
Find the voltage across each of the capacitors in the circuit shown below:



Answer: **v_1 = 30V v_2 = 30V v_3 = 10V v_4 = 20V**

Inductors

- An inductor is a passive element designed to store energy in its magnetic field.
- An inductor consists of a coil of conducting wire.



$$L = \frac{N^2 \mu A}{l}$$

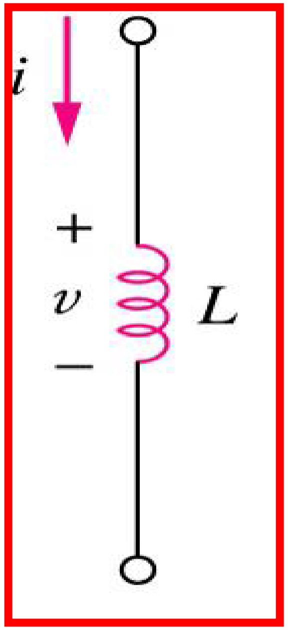
$$v = L \frac{di}{dt}$$

- Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).
- An inductor acts like a short circuit to dc (di/dt = 0) and its current cannot change abruptly.

Inductors Voltage and Current

- The current-voltage relationship of an inductor:

$$i = \frac{1}{L}\int_{t_0}^{t} v(t)\,d\,t + i(t_0)$$



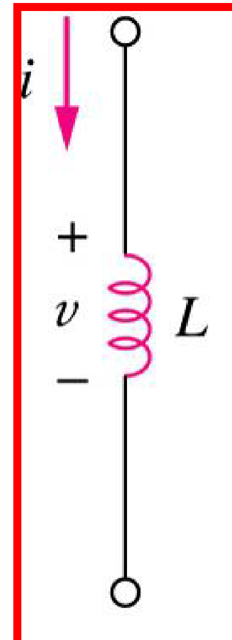
- The energy stored by an inductor:

$$w = \frac{1}{2} L\, i^2$$

Inductors

Example 5

The terminal voltage of a 2-H inductor is $v = 10(1-t)$ V Find the current flowing through it at $t = 4$ s and the energy stored in it within $0 < t < 4$ s. Assume $i(0) = 2$ A.

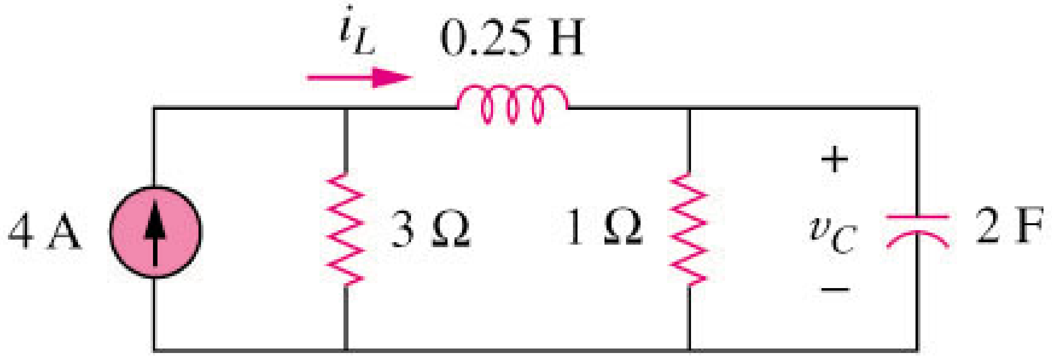


Answer: i(4s) = -18A   w(4s) = 320J

Inductors

Example 6

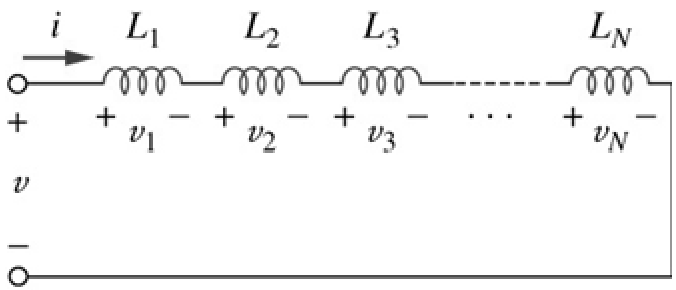
Determine v_c, i_L, and the energy stored in the capacitor and inductor in the circuit shown below under dc conditions.

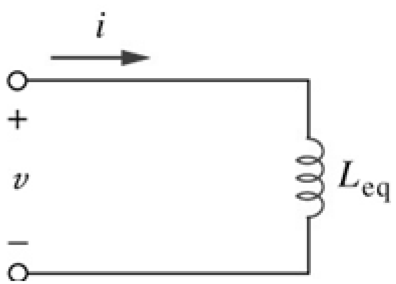


Answer: $i_L = 3A \quad v_C = 3V \quad w_L = 1.125J \quad w_C = 9J$

Series Inductors

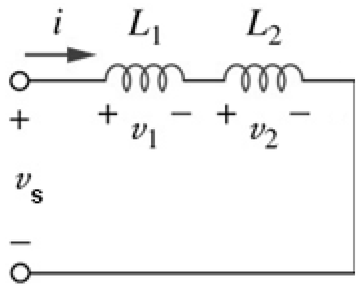
- The equivalent inductance of **series-connected** inductors is the sum of the individual inductances.





$$L_{eq} = L_1 + L_2 + \ldots + L_N$$

- Voltage Divider Rule

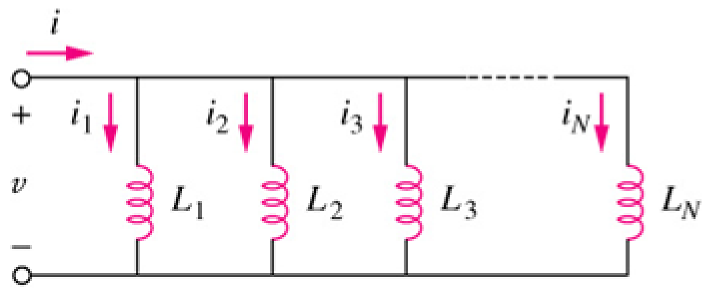


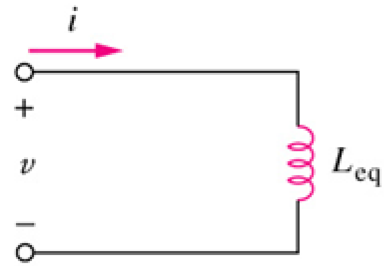
$$v_1 = \frac{L_1}{L_1 + L_2} v_s$$

$$v_2 = \frac{L_2}{L_1 + L_2} v_s$$

Parallel Inductors

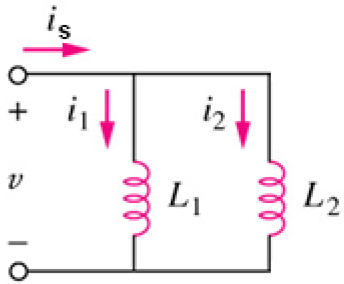
- The equivalent capacitance of **parallel** inductors is the reciprocal of the sum of the reciprocals of the individual inductances.





$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$$

- Current Divider Rule



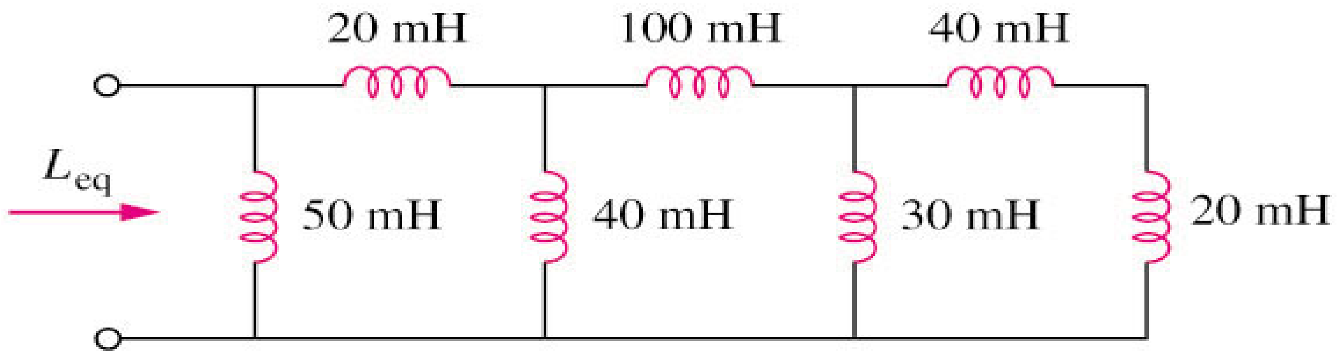
$$i_1 = \frac{L_2}{L_1 + L_2} i_s$$

$$i_2 = \frac{L_1}{L_1 + L_2} i_s$$

Series and Parallel Inductor

<u>Example 7</u>

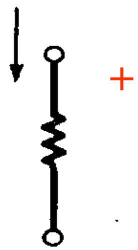
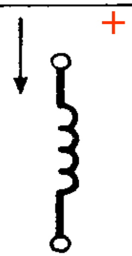
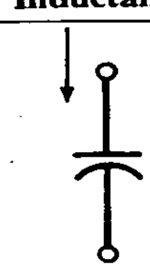
Calculate the equivalent inductance for the inductive ladder network in the circuit shown below:



Answer: **L_{eq} = <u>25mH</u>**

Current and voltage relationship

- Current and voltage relationship for R, L, C

Circuit element	Units	Voltage	Current	Power
 **Resistance**	ohms (Ω)	$v = Ri$ **(Ohm's law)**	$i = \dfrac{v}{R}$	$p = vi = i^2R$
 **Inductance**	henries (H)	$v = L\dfrac{di}{dt}$	$i = \dfrac{1}{L}\int v\,dt + k_1$	$p = vi = Li\dfrac{di}{dt}$
 **Capacitance**	farads (F)	$v = \dfrac{1}{C}\int i\,dt + k_2$	$i = C\dfrac{dv}{dt}$	$p = vi = Cv\dfrac{dv}{dt}$